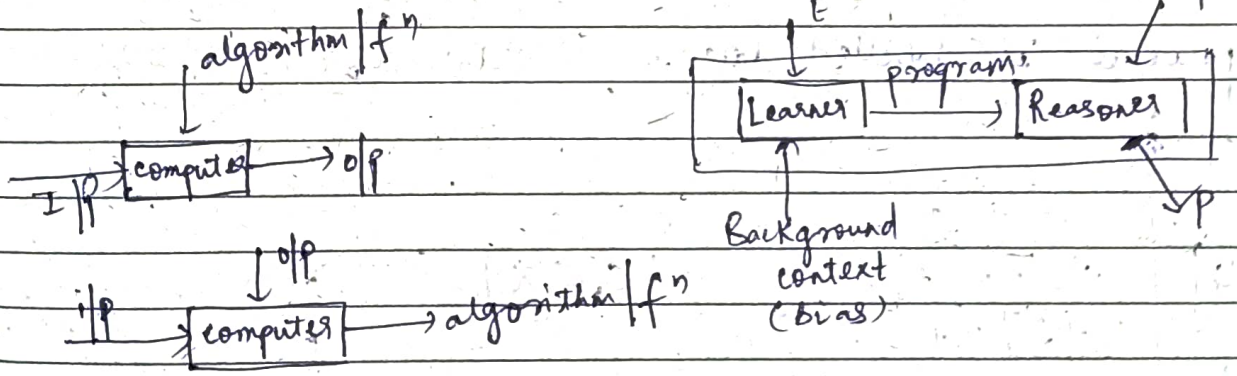
MLFSA EEL505

Introduction
1. Linear Regression, Decision trees
2. Instance based learning
3. Probability & Bayes Learning
4. Support vector machine
5. Neural Network
6. Clustering & computational concepts

Intro
History: def^n types
Space, Bias
Evaluation, sets and validation

→ Learning is ability to increase one's behaviour with experience
exploring algorithm to build model for prediction, prediction or decision making



→ A computer program is said to learn from experience 'E' wrt some class of task 'T' & performance matrix 'P' if its performance on the task 'T' improves as measured by 'P' improves with experience 'E'.

'E'

→ choose training data
→ choose a target f^n ('T')
→ How to represent your target f^n
→ choose a learning algo. to infer target f^n

Types of ML

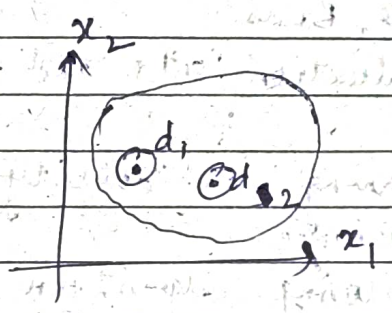
1) Supervised learning

$E \Rightarrow [x, y]$ $\quad x \rightarrow$ i/p feature

$y = f(x)$ $\quad y \rightarrow$ target feature

$f(x) \rightarrow$ target f^n

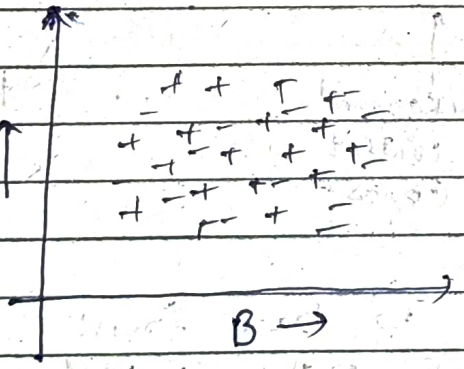
Instance	x	y
1	$x_{11}, x_{12} \ldots x_{1n}$	y_1
2	$x_{21}, x_{22} \ldots x_{2n}$	y_2
⋮	⋮	⋮
m	$x_{m1}, x_{m2} \ldots x_{mn}$	y_m



d_1 $x_1 = 2, x_2 = 5$

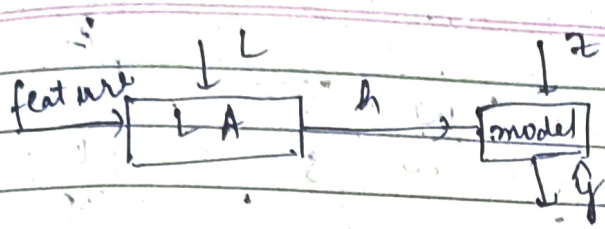
d_2 $x_1 = 4, x_2 = 3$

experience is labelled data.



$C \subset X$

Given language, feature set, set of hypothesis — All possible hypothesis call legal target f^n given E, L)



2) Unsupervised learning

3) Semi supervised learning

4) Reinforcement learning

INPUT FEATURES

1) categorical → A, B, AB, O
2) Ordinal → high, low, tall, short
3) Integer
4) Real value

O/P FEATURES

1) discrete (classifying)
2) continuous (regression)

	x_1	x_2	x	x_3	x_4
Instance	Author	Thread	Action	Situation	Type
E_1	K	N	S'	H	Long
E_2	U	N	Skip	W	Short
E_3	U	O	R'	R	L
E_4	K	O	S'	W	L
E_5	K	N	R'	H	S
E_6	K	O	R'	H	L

Concept C	Feature
$C \subseteq X$	Feature space
	Instance X

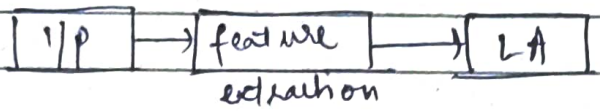
K → Known
U → unKnown
S' → skip
R' → read
L → long
S → short
H → home

$H \rightarrow h$, $x \rightarrow y$

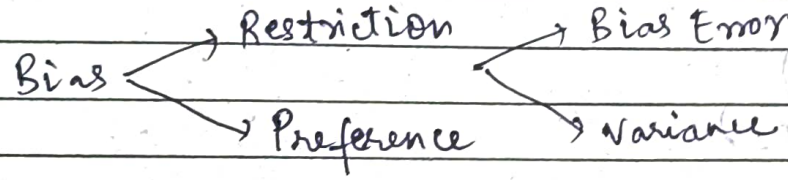
E7	K	N	W	S	
E8	V	N	W	S	

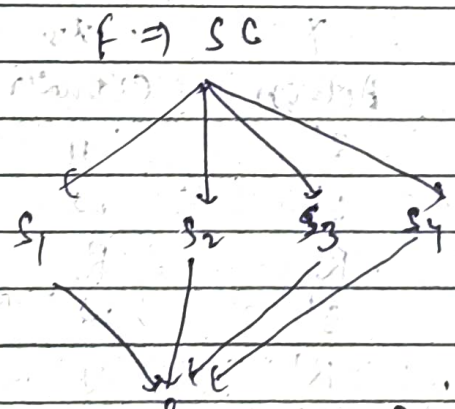
x_5
E
E
E
F
F
F

Noise
Bias



- $y \rightarrow$ boolean features
no of instance possible $\rightarrow X = 2^N = 16$
how many target $f^n = 2^{2^N} = H$





→ Generalized
→ Test instance
→ Consistent

underfitting → heavy bias, simple h, missing intricacies
low training accuracy, low Testing accuracy.
overfitting → high training accuracy,
low testing accuracy

Performance metrics - E, T, P

H, S $\xrightarrow{\text{choose}}$ h

How good is h?

$h(x) \rightarrow \hat{y}$

$e = (\hat{y} - y)$

absolute error $= |\hat{y} - y|$

E

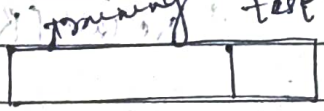
↓

dataset

Instance - x

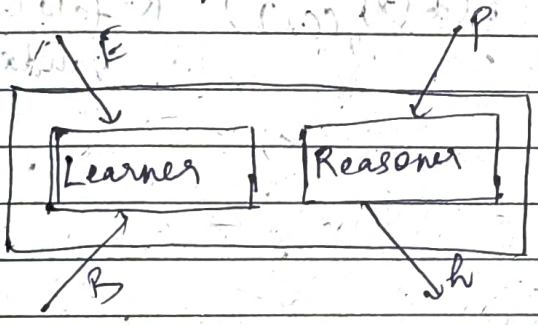
feature - feature space description

Accuracy
Error
Precision
Recall
confusion matrix



$y = f(x)$

↓ concept



→ Sum of square error

$$SSE = \sum_{i=1}^{N} (\hat{y}_i - y_i)^2$$

Avg. absolute error $= \frac{1}{N} \sum_{i=1}^{N} |\hat{y}_i - y_i|$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (\hat{y}_i - y_i)^2}$$

avg. no of misclassification $= \frac{1}{N} \sum_{i=1}^{N} g_i(\hat{y}, y)$

where $g_i(\hat{y} = y) = 1$ if $\hat{y} = y$

0 otherwise

hypothesized class \ True class	Pos	neg
Pos	TP	FP
neg	FN	TN

HC \ TC	R	G	B
R	TP	FP/FN	FP/FN
G	FP/FN	TP	FP/FN
B	FP/FN	FP/FN	TP

$$\text{Precision}(P) = \frac{TP}{TP + FP}$$

$$\text{Recall}(R) = \frac{TP}{TP + FN}$$

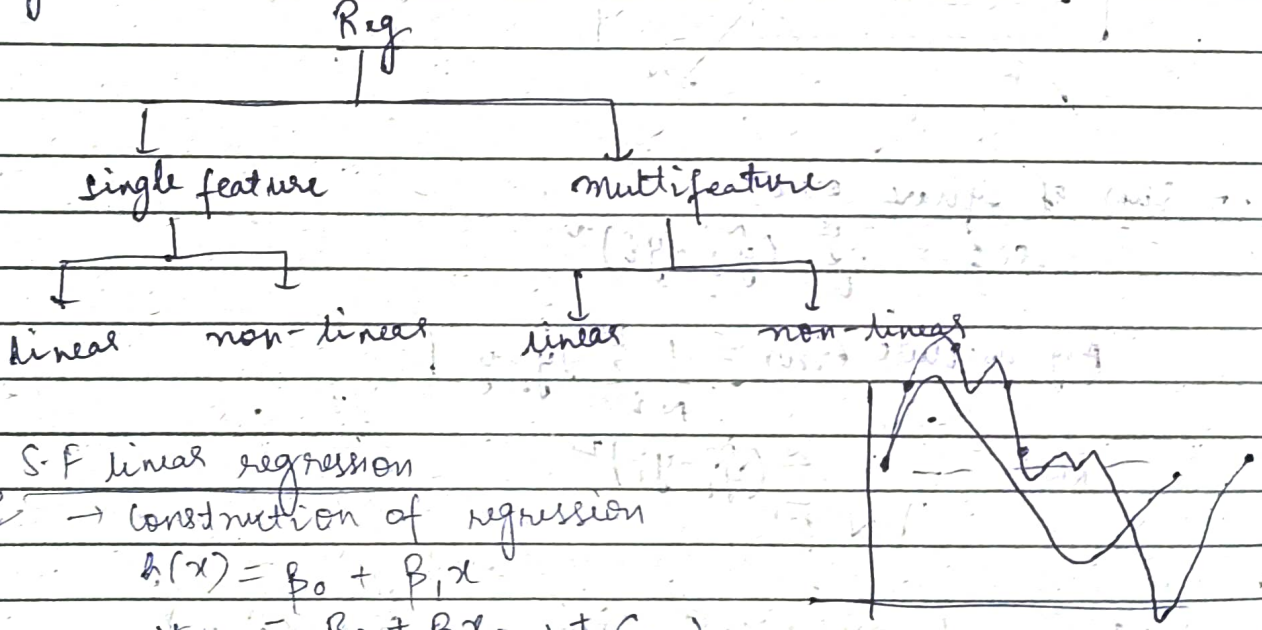
F1 score = Harmonic mean(P, R)

Sample error & true error

$$\text{Sample error} = \frac{1}{N} \sum_{x \in T} \delta(h(x), c(x))$$

K-fold cross validation

Regression



→ S.F linear regression

10/8/25 → construction of regression

$$h(x) = \beta_0 + \beta_1 x$$

$$y_{(k)} = \beta_0 + \beta_1 x_{(k)} + \varepsilon_{(k)}$$

$$y_1 = \beta_0 + \beta_1 x_{(1)} + \varepsilon_{(1)}$$

$$y_2 = \beta_0 + \beta_1 x_{(2)} + \varepsilon_{(2)}$$

Noise model

$E(\varepsilon) = 0$

$V(\varepsilon) = \sigma^2$

In pdf → Gaussian distribution
then noise model is white Noise

Parameters | weight | estimation

$\theta = [\beta_0, \beta_1]$

$J(\theta) = \frac{1}{2}[y - h(x)]^2$

$$\min J(\theta) = \frac{1}{2}\left[y_{(k)} - (\beta_0 + \beta_1 x_{(k)})\right]^2$$

$$\beta_1 = \frac{p \sum_{k=1}^{p} x_{(k)} y_{(k)} - \sum_{k=1}^{p} x_{(k)} \sum_{k=1}^{p} y(k)}{p \sum_{k=1}^{p} x_{(k)}^2 - \left(\sum_{k=1}^{p} x_{(k)}\right)^2}$$

$$\beta_0 = \frac{\sum_{k=1}^{p} y(k) - \beta_1 \sum_{k=1}^{p} x_{(k)}}{p}$$

p : no of training samples

Multi feature linear regression

$[x], x_1, x_2 \ldots x_N$

N : total no of features

$h(\bar{x}) = \beta_0 x_N + \beta_1 x_1 + \beta_2 x_2 \ldots \beta_N x_N$

$= \beta_0 + \sum_{j=1} \beta_j x_j$

need N no of estimators for actual target fn

N no of predictor variable

→ To find analytically involve N×N matrix inversion hence we heavily rely on iterative method which update β_j at each iteration.

LMS → least minimum slope

cost f^n, J(θ) → Quadratic f^n
→ convex f^n

Delta rule

single training pt

$$J(\theta) = \frac{1}{2} \sum_{i=1}^{m} [h(\bar{x})^{(i)} - y^{(i)}]^2$$

find θ that minimize J(θ)

→ start search with initial guess of $\theta [\beta_0^{(k)}, \beta_1^{(k)} \ldots \beta_n^{(k)}]$

$$\frac{\partial}{\partial \theta} J(\theta) = \frac{\partial}{\partial \theta_j} \left[\frac{1}{2} h(\bar{x}) - y \right]^2$$

$$= [h(\bar{x}) - y] \frac{\partial}{\partial \beta_j} [h(\bar{x}) - y]$$

$$= [h(\bar{x}) - y] x_j$$

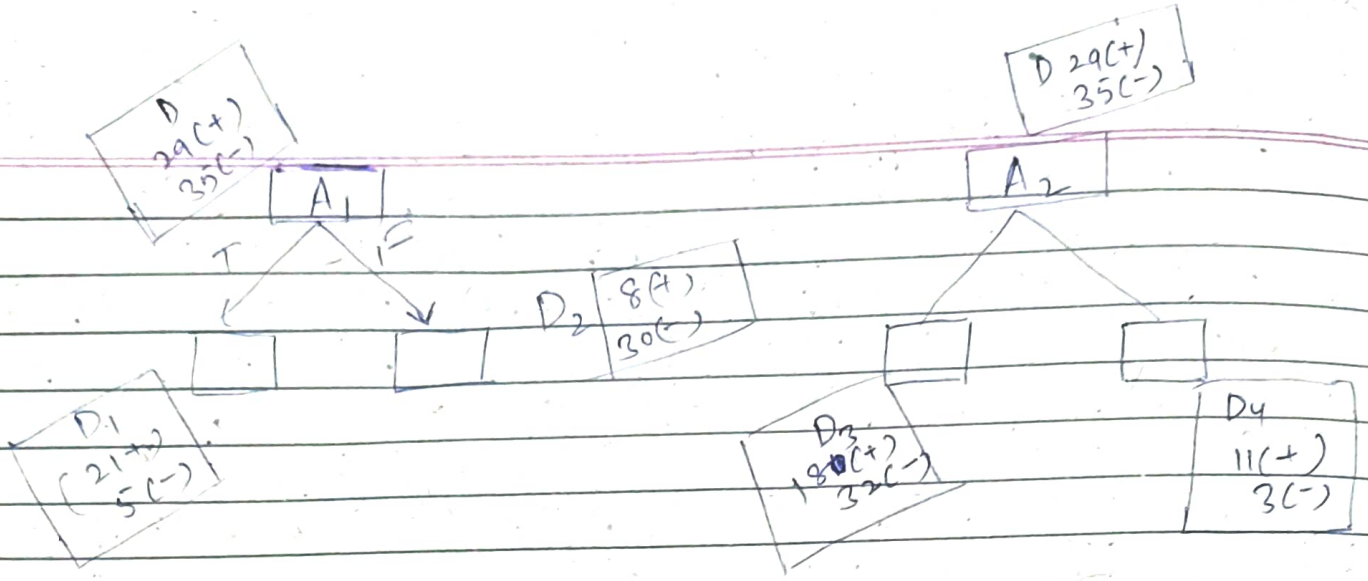
$$\beta_j^{(i+1)} = \beta_j^{(i)} + \alpha (y^{(i)} - h(\bar{x}^{(i)})) x_j^{(i)}$$

$y^{(i)} - h(\bar{x}^{(i)}) = 0$

> 0 increase β_j

< 0 decrease β_j

$$\beta_j^{(i+1)} = \beta_j^{(i)} + \alpha [y_{(k)}^{(i)} - h(\bar{x}_{(k)}^{(i)})] x_j^{(k)}$$



Entropy

$$\text{Entropy}(D) = P_+ \cdot [-\log_2(P_+)] + (P_-)[-\log_2(P_-)]$$

=

$$\text{Gain}(D, A) = S(D) - \sum_{v \in \text{values}(A)} \frac{|D_v|}{|D|} S(D_v)$$

$S(D) = 0.99$, $S(D_1) = 0.7$, $S(D_2) = 0.74$, $S(D_3) = 0.94$,
$S(D_4) = 0.75$

Instances	Weather	Temperature	Humid	Wind	Action
1	S	H	H	W	X
2	S	H	H	S	X
3	C	H	H	W	✓
4	R	M	H	W	✓
5	R	C	N	W	✓
6	R	C	N	S	X
7	C	C	N	S	✓
8	S	M	H	W	X
9	S	C	N	W	✓
10	R	M	N	W	✓
11	S	M	N	S	✓
12	C	M	H	S	✓
13	C	H	N	W	✓
14	R	M	H	S	X
15	S	C	H	W	X
16	C	C	N	W	✓

$$Gain(P, A_1) = S(D) - \left[\frac{D_1}{D} S(D_1) + \frac{D_2}{D} S(D_2)\right]$$

$$= 0.99 - \left[\frac{26}{64} 0.71 + \frac{38}{64} (0.74)\right]$$

$$= 0.99 - [0.51 + 0.20]$$

$$= 0.262$$

$Gain(D, A_2) = 0.09$

$S(D) = 0.95$

$G(D, \text{weather}) =$

$G(D, \text{temp})$

$G(D, \text{hum}) = 0.20$

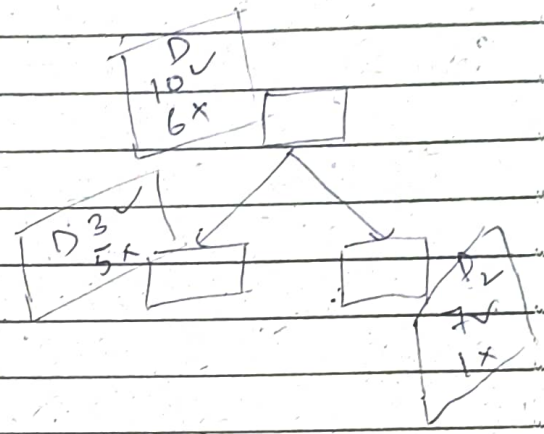
$G(D, \text{wind}) = 0.30$

$S(D) = 0.95$

$S(D_1) = 0.54$

$S(D_3) = 0.88$

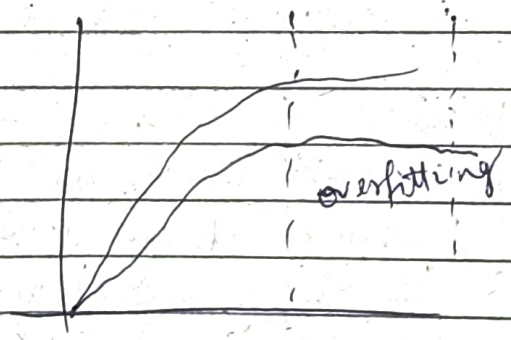
$S(D_4) = 1.0$



Overfitting

→ A hypothesis 'h' is set to overfit a training data if there exists another hypothesis 'h' s.t. 'h' has more error than h. On training set but 'h' has less error than h on any unseen test set.

Pruning



Pre pruning / Pruning / post pruning

- reduced error post pruning

Early stop instances

(1) number of ~~thresholds~~ is less than threshold (GINI index) decided by the user.

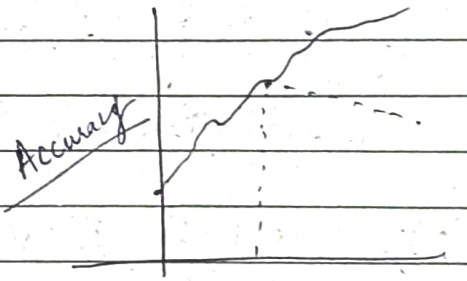
(2) The available instances are unconnected with residual attributes
→ correlation coeff.
→ χ^2 test

(3) less gain of information

~~Ten~~ Training

Grow set

validation
- X-validation
  - K fold Xvalidation



↑ train error / test error ↓ → complexity h(c)

↑ train error ↓, test error ↓ → training data size → Generalization

07/09/26

K-nearest neighbour

Instance based learning
↳ Lazy algorithm

prediction time → get a test instance → forming the model

Large feature size → feature selection / ~~extra~~ feature dimension is the problem

$\{x_i, y_i\}$

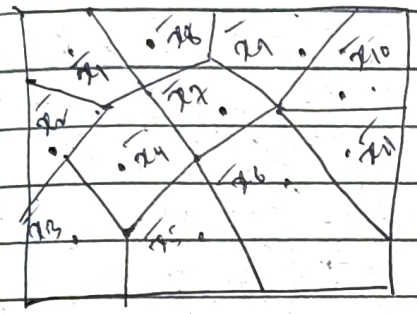
$\{x_i, f(x_i)\}$

↳ prediction time

Test instance x_t → looping for which training instance bears most similarity → with which the distance is minimum.

Eucledian distance

Voronoi Diagram



find 1st bisector of two adjacent training instances and form decision boundary.

The decision boundary

$\bar{x}_1 = [x_{1a}, x_{1b}]$

$\bar{x}_2 = [x_{2a}, x_{2b}]$

y_1 - Red
y_2 - Red
y_3 - Green
y_4 - Red
y_5 - Red
y_6 - Green
y_7 - Green
y_8 - Red
y_9 - Green
y_{10} - Green
y_{11} - Green

(1) Noise in attribute

(2) Noise in target level

Moving average

Distance = ~~$D_E(x_{ij} - x_{tj})$~~ $D_E(x_{1k}, x_t)$

Eucledian

$$= \sqrt{\sum_{j=1}^{m} (x_{1k_j} - x_{t_j})^2}$$

$\bar{x}_{1k} = [x_{1k_1}, x_{1k_2} \cdots x_{1k_m}]$ training instances

$\bar{x}_t = [x_{t_1}, x_{t_2} \ldots\ldots x_{t_m}]$ test instance

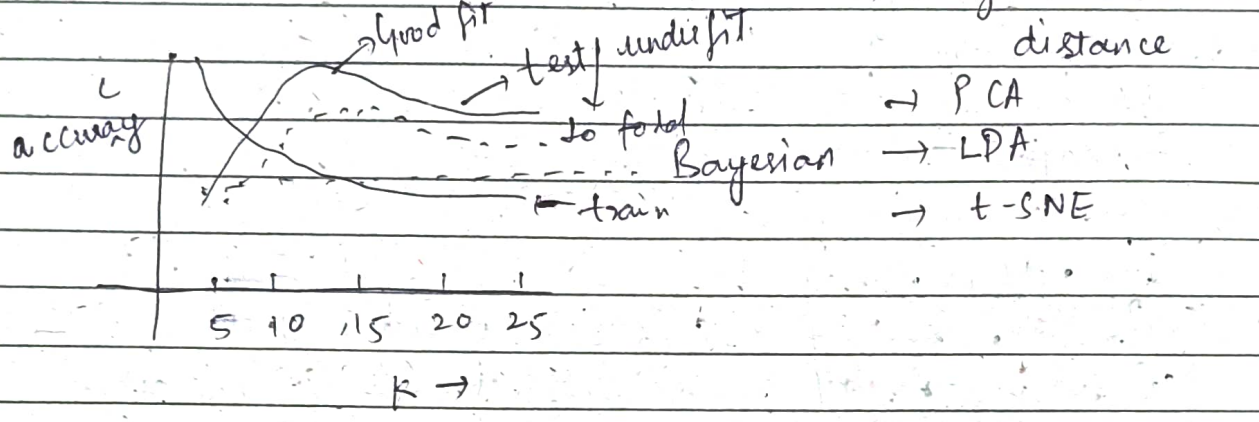
Classification

$[y_1, y_2 \ldots\ldots y_n]$ majority class

Regression

$\frac{y_1 + y_2 \ldots\ldots y_K}{K}$

→ weighted euclidean distance



→ PCA

→ LDA

→ t-SNE